

BankAccountManagementSystem

Java Programming Laboratory Experiment

Experiment No.	03
Date	11/09/2026
Title	Bank Account Management System using Classes and Objects

Objective / Aim

To design and implement a Bank Account Management System in Java using classes and objects, demonstrating data members, object creation, methods, conditional statements, deposits, withdrawals, and displaying account details.

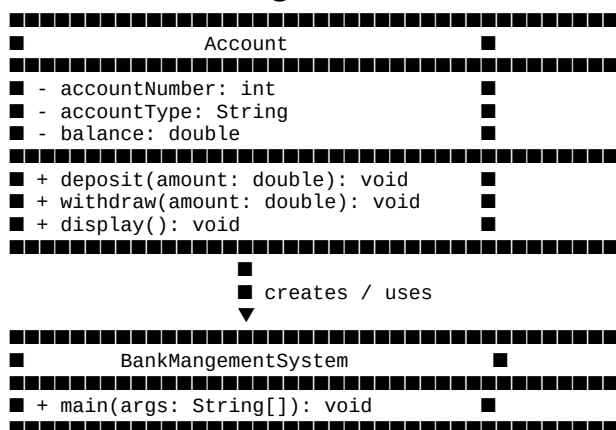
System Requirements

HARDWARE	SOFTWARE
• Standard PC/Laptop • Intel Core i3/i5 processor or equivalent • Minimum 4 GB RAM • Minimum 500 MB free storage	• Operating System: Windows / Linux / macOS • Java Development Kit (JDK 17 or JDK 21) • Text Editor / IDE: Eclipse, IntelliJ IDEA, NetBeans, or VS Code • Command Prompt / Terminal

Algorithm / Logical Description

1. Create an Account class.
2. Declare account number, account type, and balance as data members.
3. Define deposit() to add money to the account balance.
4. Define withdraw() to check available balance and deduct the withdrawal amount when sufficient balance is available.
5. Define display() to show account details.
6. Create an Account object in main().
7. Assign account number, account type, and initial balance.
8. Display the initial account details.
9. Perform deposit and withdrawal transactions.
10. Display the updated account details after transactions.

UML Class Diagram



Relationship: The BankMangementSystem class creates and uses an object of the Account class. The Account class contains data members and methods for managing account transactions.

Source Code

```
// Account class represents a bank account
class Account {

    // Data members of the Account class
    int accountNumber;
    String accountType;
    double balance;
```

```

// Method to deposit money into the account
void deposit(double amount) {

    // Adding deposit amount to the balance
    balance = balance + amount;

    // Displaying deposited amount
    System.out.println("Deposited: " + amount);
}

// Method to withdraw money from the account
void withdraw(double amount) {

    // Checking whether sufficient balance is available
    if (amount <= balance) {

        // Deducting withdrawal amount from balance
        balance = balance - amount;

        // Displaying withdrawn amount
        System.out.println("Withdrawn: " + amount);

    } else {

        // Displaying message when balance is insufficient
        System.out.println("Insufficient balance");
    }
}

// Method to display account details
void display() {

    // Displaying account number
    System.out.println("Account Number: " + accountNumber);

    // Displaying account type
    System.out.println("Account Type: " + accountType);

    // Displaying account balance
    System.out.println("Balance: " + balance);
}

}

// Main class of the Bank Management System
public class BankManagementSystem {

    // Main method - program execution starts here
    public static void main(String[] args) {

        // Creating an object of Account class
        Account a = new Account();

        // Assigning account number
        a.accountNumber = 101;

        // Assigning account type
        a.accountType = "Savings";

        // Assigning initial account balance
        a.balance = 5000;

        // Displaying initial account details
        a.display();
        // Depositing Rs.2000 into the account
        a.deposit(2000);
    }
}

```

```
// Withdrawing Rs.1000 from the account
a.withdraw(1000);

// Displaying account details after transactions
System.out.println("\nAfter Transactions:");
a.display();
    }
}
```

Completion and Execution Commands

Save the source code exactly as:

BankMangementSystem.java

Compile the program:

javac BankMangementSystem.java

Run the program:

java BankMangementSystem

Output Screen

Account Number: 101
Account Type: Savings
Balance: 5000.0
Deposited: 2000.0
Withdrawn: 1000.0

After Transactions:
Account Number: 101
Account Type: Savings
Balance: 6000.0

Results

The Bank Account Management System was successfully implemented in Java using classes and objects. The program successfully performs deposit and withdrawal operations and displays the account details before and after transactions.